Wei Wang

Department of Agricultural, Environmental, and Development Economics | The Ohio State University
2120 Fyffe Road, Columbus, OH 43210 | wang.16091@osu.edu | (+1) 614-589-9657 | weiwang-econ.github.io

Education

The Ohio State University — Columbus, OH
Ph.D. in Agricultural, Environmental, and Development Economics — Aug. 2022 – May 2027 (expected)
Advisor: Yongyang Cai

Xiamen University — Xiamen, China
M.S. in Quantitative Economics — Sep. 2017 – Jun. 2020

Jiangxi University of Finance and Economics — Nanchang, China
B.S. in Mathematical Economics — Sep. 2013 – Jul. 2017

Research Fields and Methods

Primary Fields: Environmental and Energy Economics; Climate Economics; International Trade
Additional Fields: Food and Applied Behavioral Economics; Nonmarket Valuation
Methods: Dynamic Integrated Assessment Modeling; Quantitative General Equilibrium Modeling; Applied Econometrics; Discrete Choice Experiments

Job Market Paper

"Solar Tariffs and the Energy Transition: Clean-Investment Dynamics and Climate Policy Responses in the Global Economy."
Trade protection can support domestic clean-energy manufacturing but also raise deployment costs and delay the energy transition. To study whether endogenous climate policy can offset these costs, I develop a multi-region, forward-looking integrated assessment model with bilateral trade in clean-energy investment goods, endogenous renewable and fossil generation capacity, and strategic climate policy. I also develop an algorithm that jointly solves goods-market and open-loop Nash equilibria. Applied to U.S. solar tariffs, the model shows that tariffs expand domestic module production but slow clean-capacity accumulation and raise U.S. and global emissions. The optimal U.S. carbon tax may rise but cannot eliminate investment and trade distortions. U.S. welfare is hump-shaped: moderate protection may be beneficial, while higher tariffs ultimately worsen climate and welfare outcomes.

Publications and Accepted Papers

"Evaluating the Optimal Air Pollution Reduction Rate: Evidence from the Transmission Mechanism of Air Pollution Effects on Public Subjective Well-Being." (Corresponding author; with Chuanwang Sun, Xiangyu Yi, Tiemeng Ma, and Weiyi Cai). *Energy Policy*, 2022.

"Expanding Firms' Innovation Boundaries: The Role of Suppliers' Breakthrough Green Innovation and Digitalization." (with Mingming Zhao, Fuxiang Wu, and Xia Xu). *IEEE Transactions on Engineering Management*, accepted.

Working Papers

"The Long-Term Effects of the EU Carbon Border Adjustment Mechanism." (with Yongyang Cai).

"Rising Temperatures, Structural Transformation, and Within-Country Wage Inequality."

"Throw a Bunch Away: A Bundled Utility Maximization Framework Modeling Multiple-Item Food Discarding." (with Wuyang Hu, Xiaolei Li, Jian Li, and Ping Qing).

Work in Progress

"Non-Cooperative Climate Policy in a Dynamic Trade IAM." (with Yongyang Cai).

"Capital Flows and Optimal Climate Policies." (with Jingke Wu, Xufeng Liu, and Yongyang Cai).

Conference and Workshop Presentations

Agricultural and Applied Economics Association Annual Meeting	2026
AERE@WEAI Annual Conference	2026
Berkeley/Sloan Summer School in Environmental and Energy Economics	2025
Southern Agricultural Economics Association Annual Meeting	2025
NAREA Scholars' Circle	2024

Research Experience

Graduate Research Assistant, The Ohio State University — Spring 2025 – Summer 2025
USDA-funded research on the EU Carbon Border Adjustment Mechanism; supervisor: Yongyang Cai

Teaching Experience

Lab Instructor, The Ohio State University

Advanced Quantitative Methods I — Autumn 2025
Student Evaluation of Instruction: **4.9/5**.

Advanced Quantitative Methods II — Autumn 2025
Student Evaluation of Instruction: **4.9/5**.

Teaching Associate, The Ohio State University

Principles of Food and Resource Microeconomics	Spring 2026
Energy, the Environment, and the Economy	Autumn 2024
Economics of Public Policy Analysis	Spring 2024
Strategic Management	Autumn 2022 and Autumn 2023

Awards and Grants

Berkeley/Sloan Summer School Travel Grant	2025
AEDE Travel Grant, The Ohio State University	2025 – 2026
First Prize, China National College Students Competition on Energy Economics	2018
Outstanding Graduate Award, Jiangxi University of Finance and Economics	2017
National Scholarship, Ministry of Education of China	2016

Professional Service

Conference Discussant, AERE@WEAI Annual Conference	2026

Skills and Languages

Software: GAMS, MATLAB, Stata, R, Python, ArcGIS, LaTeX
Languages: Chinese (native), English (fluent)